

Infections

Herpes and zoster, tuberculosis, norovirus and rotavirus (FUT2), CCR5-Δ32, HPV and warts, sepsis, hepatitis B and C and innate defenses

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The variants with the greatest effect on specific viruses are on the good side. **IFNL4 (IL28B)** is C/C in rs12979860 and T/T in rs8099917, the genotype with the most spontaneous elimination of hepatitis C. **HLA-DPB1 (A/A)** and **HLA-DQ (G/G)** are on the side of elimination of hepatitis B virus. The **OAS1** has 1 G allele (prenylated antiviral form) and the COVID-19 severe is at percentile 4.7. There is no **HLA-B*57:01** (rs2395029 T/T), so there is no risk of hypersensitivity to abacavir. TLR4 D299G, TLR2 R753Q and MBL2 G54D are absent. The loci loads of HIV (33.6), warts and HPV (29.3), sepsis (36.5) and pneumonia and influenza (36.2) are below the median.

Unfavourable. The tuberculosis loci load is at 94.0, with the risk allele of **ASAP1** in double dose (A/A). The loci load of hepatitis B is at 88.2, despite the two large positive loci HLA. Herpes zoster is at percentile 74.9 and urinary infection in 93.8 (already in §103). Sérgio F is secretory (FUT2 G/A), so it is susceptible to the norovirus GII.4 and rotavirus, like most people. It does not have **CCR5-Δ32** (+/+), so it does not have natural protection against HIV R5.

What does it mean? The classical antiviral defense (lambda interferons, OAS1, HLA of hepatitis) is good. The risk lies in the polygenic signs of tuberculosis and herpes zoster, which only weigh with exposure or immunosuppression. The most successful measures are vaccines (zone, hepatitis B confirmed by anti-HBs, influenza, COVID-19 and pneumococcal, §103) and a latent tuberculosis test before any immunosuppressive. This report complements §103, which treated pneumococci and bacterial infections.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 14 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 of the chip, 1 imputed and 1 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 4,412,143 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 250 loci, 168 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In all these scores, a higher percentile = more risk. Herpes zoster, urinary infection and severe COVID-19 maintain the values of §103. The genomic scores of warts, hepatitis B and cellulitis are new.

Herpes and warts

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Herpes zoster	PGS001131 (Tanigawa 2022, 82 variants)	62 81 (weighted coverage 66%)	74.9	TYPICAL	typical
Viral warts (HPV, genomic)	PGS018443 (Chen 2025, PRS-CS, 984 thousand variants)	950,708 983,817 (weighted coverage 96%)	44.2	TYPICAL	typical

Hepatitis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Viral hepatitis B (genomic)	PGS018427 (Chen 2025, PRS-CS, 984 thousand variants)	950,654 983,762 (weighted coverage 96%)	56.4	TYPICAL	typical

Bacteria and sepsis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cellulitis and skin abscesses	PGS002081 (Privé 2022, LDpred2, 641 thousand variants)	628,084 640,921 (weighted coverage 97%)	48.8	TYPICAL	typical
Urinary infection (genomic)	PGS019461 (Chen 2025, PRS-CS, 984 thousand variants)	950,715 983,825 (weighted coverage 96%)	93.8	MORE LIKELY	unfavourable

- **Urinary infection (genomic):** In §103 (Pneumococcus).

Respiratory viruses

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Severe COVID-19 (genomic)	PGS004938 (Kovalenko 2024, LDpred2, 956 thousand variants)	931,920 955,503 (weighted coverage 97%)	4.7	LESS LIKELY	favourable

- **Severe COVID-19 (genomic):** Already in a previous report.

4. Key variants by topic

Variants of known effect on viruses and specific mycobacteria. The CCR5-Δ32 (deletion) was read in the imputation (DR2 = 1.00). IFITM3 rs12252 did not pass the imputation quality threshold, as in §44.

Norovirus, rotavirus and blood groups

Mucosal and blood antigens that enteric viruses use as receptors.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
FUT2	rs601338	W154X	G/A	chip	A (1 copy/s)	typical	W154X: 2 copies of A = non-secretor, resistant to norovirus GII.4 and rotavirus P[8]; secretors (at least 1 G) are susceptible	Strong
ABO	rs505922		T/T	chip	C (0 copy/s)	without judgment	The C allele marks the groups A/B; T/T suggests group O (more susceptible to some noroviruses and severe forms of cholera; less thrombosis)	Moderate

HIV and CCR5

HIV correceptor and the HLA allele that controls the virus (and predicts abacavir allergy).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HCP5 (HLA-B*57:01)	rs2395029	B*57:01	T/T	chip	G (0 copy/s)	favourable	HLA-B*57:01 marker: spontaneous control of HIV and, above all, hypersensitivity to abacavir	Strong
CCR5	CCR5-Δ32	Δ32 (deletion of 32 pb)	+/+ (without Δ32)	imputation DR2=1.00	Δ32 (0 copy/s)	without judgment	2 copies of Δ32 block the entry of HIV R5; 1 copy delays the progression. Δ32 is associated with worse evolution in some flavivirus infections	Strong

Hepatitis B and C

Lambda (IFNL4) and HLA class II: spontaneous elimination of hepatitis viruses.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IFNL4 (IL28B)	rs12979860		C/C	chip	C (2 copy/s)	favourable	The C allele (C/C) duplicates the spontaneous elimination of hepatitis C virus	Strong
IFNL4 (IL28B)	rs8099917		T/T	chip	T (2 copy/s)	favourable	The T allele is associated with more spontaneous elimination of hepatitis C	Strong
HLA-DPB1	rs9277535		A/A	chip	G (0 copy/s)	favourable	HLA-DP region: G allele is associated with persistence of chronic hepatitis B and A elimination of the virus (strong and replicated effect)	Strong
HLA-DQ	rs7453920		G/G	chip	A (0 copy/s)	favourable	HLA-DQ region: persistence of chronic hepatitis B [GWAS Catalog: Chronic hepatitis B infection, allele A, OR 2.00, p=1E-60, PMID 25802187]	Moderate

Innate immunity (TLR, MBL, interferons)

Microbes sensors (TLR), mannose binding lectin and interferon-induced antiviral proteins.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TLR4	rs4986790	D299G	A/A	chip	G (0 copy/s)	favourable	D299G: TLR4 less responsive to LPS; inconsistent associations with Gram-negative sepsis	Weak
TLR2	rs5743708	R753Q	G/G	chip	A (0 copy/s)	favourable	R753Q (rare): less TLR2 signal; tuberculosis and staphylococcal infections	Weak
MBL2	rs1800450	G54D	C/C	chip	T (0 copy/s)	favourable	G54D (B allele): low mannose binding lectin; more childhood infections and immunosuppression	Moderate
IFITM3	rs12252	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	SNP not found in the data	C. copy/s)	without data	The C allele creates a truncated IFITM3; more severe flu (especially in C/C carriers)	Weak
OAS1	rs10774671	G/A	G/A	chip	G (1 copy/s)	favourable	The G allele keeps the OAS1 prenylated, antiviral; Less severe COVID-19	Moderate

Herpes, tuberculosis, warts and sepsis (GWAS)

Replicated tuberculosis susceptibility locus.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
Tuberculosis (ASAP1)	rs4733781		A/A	chip	A (2 copy/s)	unfavourable	ASAP1 of dendritic cells: migration and tuberculosis [GWAS Catalog: Tuberculosis, allele A, p=4E-12, PMID 25774636]	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 510-8$) by infection. Alleles are oriented towards more risk and the sum is compared with the reference population. Several sets have few loci; they read as trend.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Herpes (HSV, zoster, varicella)	15 (8)	10 of 16	8	61.8	typical
Tuberculosis	32 (18)	20 of 36	17	94.0	unfavourable
Hepatitis B	30 (26)	26 of 52	26	88.2	unfavourable
Hepatitis C	24 (19)	14 of 38	19	71.4	typical
HIV (acquisition and viral load)	49 (38)	28 of 76	38	33.6	typical
Warts and HPV	12 (11)	11 of 22	11	29.3	typical
Sepsis	45 (27)	8 of 54	27	36.5	typical
Pneumonia and influenza	43 (21)	17 of 42	21	36.2	typical

6. Favourable

- **Hepatitis C:** IFNL4 rs12979860 C/C and rs8099917 T/T, the genotype with ~2× more spontaneous elimination and better response to treatment. **Strong.**
- **Hepatitis B:** HLA-DPB1 rs9277535 A/A and HLA-DQ rs7453920 G/G, on the side of virus elimination and good vaccine response. Strong (locus effect), moderate in the set.
- **COVID-19 grave** at percentile 4.7; **OAS1 rs10774671 G/A** (1 antiviral form). **Moderate.**
- **HLA-B*57:01 absent** (rs2395029 T/T): no risk of hypersensitivity to abacavir. **Strong.**
- **TLR4 D299G A/A, TLR2 R753Q G/G** and **MBL2 G54D C/C**: normal sensors and lectin (coherent with §103). **Moderate.**
- **Loci loads below the median:** HIV 33.6, warts and HPV 29.3, sepsis 36.5 (45 loci; replaces the reading of 7 loci of §103, which gave 3.2), pneumonia and influenza 36.2. Warts (genomic) at 44.2 and cellulitis at 48.8, typical. **Weak.**

7. Unfavourable

- **Tuberculosis:** loci load at 94.0; **ASAP1 rs4733781 A/A** (2 risk alleles; lower migration of dendritic cells). It only weighs with exposure. **Moderate.**
- **Hepatitis B:** loci load at 88.2, despite the large favourable HLA loci; genomic score in 56.4 (typical). In practice, the vaccine eliminates this risk. **Weak.**
- **Herpes zoster** at percentile 74.9; herpes loci load at 61.8. **Weak moderate.**
- **Urinary infection** at percentile 93.8 (already in §103; it is added the effect of dapagliflozin, which puts glucose in the urine). **Moderate.**
- **Secretor (FUT2 rs601338 G/A)** and probable group O (ABO rs505922 T/T): susceptible to norovirus GII.4 and rotavirus, and group O is associated with more severe forms of some noroviruses and cholera. **Moderate.**
- **No CCR5-Δ32 (+/+):** without natural protection against HIV R5, as the vast majority of people. Strong (genotype), without practical impact without exposure.

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Immunosenescence	Antiviral defense innate good (IFNL4, OAS1, TLR, MBL); zoster and tuberculosis reactivate when cellular immunity ages	Conditional
Loss of T Naive Cells	Vaccination maintains memory against zoster and hepatitis, compensating the loss of naive T cells	Conditional
Chronic Inflammation	Chronic infections (hepatitis B and C, herpesviruses) feed low-grade inflammation; good genetics of hepatitis elimination	Favourable
Inflamming	COVID-19 severe at percentile 4.7 and sepsis with low loci load: lower trend for inflammatory storms	Favourable
Dysbiosis	Secretor (FUT2): microbiota with more bifidobacteria, but susceptible to norovirus and rotavirus	Neutral
Intestinal Barrier dysfunction	Enteric viruses bind to the antigens ABO/FUT2 of the intestinal epithelium	Neutral
Retrotransposing activation	The same antiviral sensors (OAS1, interferons) control the endogenous retroelements; OAS1 G/A intermediate	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Recombinant vaccine against the zoster** (2 doses), recommended from the 50 years, or before in those who have immunosuppression. With the zoster at the 75th percentile, it makes sense not to postpone. *Suggestion (not applied)*.
2. **Hepatitis B**: confirm the vaccination with an anti-HBs analysis. If it is < 10 mUI/mL and there is no record of complete vaccination, vaccinate (3 doses or 2 doses of the vaccine with adjuvant). *Suggestion (not applied)*.
3. **Latent tuberculosis test (IGRA) before any immunosuppression**: biological, prolonged corticosteroids or mTOR inhibitors in immunosuppressive dose. Also after close contact with a case of tuberculosis or long trips to areas of high incidence. With the loci load at percentile 94 and ASAP1 A/A percentile, it is the most actionable point. *Suggestion (not applied)*.
4. **Annual flu, COVID-19 according to pneumococcal norms and vaccines** (see §103). *Suggestion (not applied)*.
5. **Norovirus**: hand washing with water and soap (alcohol-gel is ineffective against norovirus) and care with raw shellfish. *Suggestion (not applied)*.
6. **Urinary infection**: abundant hydration (already useful with dapagliflozin) and attention to symptoms. In men, a urinary infection should always be evaluated because it may indicate a prostate problem. *Suggestion (not applied)*.
7. **Information**: the HLA-B*57:01 negative should be included in the clinical record (abacavir would be safe). The absence of CCR5-Δ32 does not change in practice. *Suggestion (not applied)*.

10. Limitations

- Several sets of loci (herpes, warts, tuberculosis) have few loci and come from small studies. They are read as trend.
- The large loci of hepatitis B and tuberculosis were discovered mainly in other populations; the size of the effect may be different in the reference population.
- The IFITM3 rs12252 did not pass the quality of the imputation. The complete HLA genotype (classes I and II) is not read directly; markers are used.
- The risk of infection depends mainly on exposure, vaccination and immune status, and very little of the genes.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs, and CCR5-Δ32, come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with DR2 $\geq$ 0.8. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. The meaning of HLA-DPB1 rs9277535 was verified in the original literature.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (6 scores: zoster, warts, hepatitis B, cellulitis, urinary infection and severe COVID-19). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 8 sets (250 loci), alleles oriented by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** Ge 2009 and Thomas 2009 (IFNL4), Kamatani 2009 (HLA-DP and hepatitis B), Fellay 2007 (HLA-B*57:01), Samson 1996 (CCR5-Δ32), Zhou 2021 (OAS1), Curtis 2015 (ASAP1), Lindesmith 2003 (FUT2 and norovirus), GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Norovirus, rotavirus and blood groups	FUT2	rs601338	G/A	chip	A	1	strand +	secretor (1 functional copy arrives): susceptible to norovirus GII.4 and rotavirus, as the majority
Norovirus, rotavirus and blood groups	ABO	rs505922	T/T	chip	C	0	strand +	0 copy of the effect allele
HIV and CCR5	HCP5 (HLA-B*57:01)	rs2395029	T/T	chip	G	0	strand +	without B*57:01 marker: abacavir without this risk
HIV and CCR5	CCR5	CCR5-Δ32	+/+ (without Δ32)	imputation DR2=1.00	Δ32	0	strand +	+/+: normal CCR5 receptor; without the protection of Δ32 against HIV R5, and also without the increased risk of severe forms of Western Nile and flu described in homozygous Δ32
Hepatitis B and C	IFNL4 (IL28B)	rs12979860	C/C	chip	C	2	strand +	2 copies of the protective allele
Hepatitis B and C	IFNL4 (IL28B)	rs8099917	T/T	chip	T	2	strand +	2 copies of the protective allele
Hepatitis B and C	HLA-DPB1	rs9277535	A/A	chip	G	0	strand +	A/A: 2 copies of the allele associated with more virus elimination and less chronic hepatitis B
Hepatitis B and C	HLA-DQ	rs7453920	G/G	chip	A	0	strand +	without the risk allele
Innate immunity (TLR, MBL, interferons)	TLR4	rs4986790	A/A	chip	G	0	strand +	without the risk allele
Innate immunity (TLR, MBL, interferons)	TLR2	rs5743708	G/G	chip	A	0	strand +	without the risk allele
Innate immunity (TLR, MBL, interferons)	MBL2	rs1800450	C/C	chip	T	0	strand +	without the risk allele
Innate immunity (TLR, MBL, interferons)	IFITM3	rs12252	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	C			SNP not found in the data
Innate immunity (TLR, MBL, interferons)	OAS1	rs10774671	G/A	chip	G	1	strand +	1 copy of the protective allele
Herpes, tuberculosis, warts and sepsis (GWAS)	Tuberculosis (ASAP1)	rs4733781	A/A	chip	A	2	strand +	2 risk allele copies